VAULTMESH v2.0: ENTERPRISE CLOUD SECURITY POSTURE MANAGEMENT (CSPM) & AUTOMATED REMEDIATION PLATFORM

Course: Cloud Computing — Major Project Submission
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Live Deployment: https://rev0nix.github.io/VaultMesh/

1. Executive Summary

VaultMesh v2.0 is an enterprise-grade, cloud-native Cloud Security Posture Management (CSPM) and Security Orchestration, Automation, and Response (SOAR) platform for AWS environments. This Major Project upgrades the architecture into containerized microservices orchestrated via Docker Compose and Kubernetes (K8s), backed by PostgreSQL 15 persistence, Redis 7 message brokering, Celery asynchronous task workers, Boto3 automated remediation playbooks, and Terraform Infrastructure as Code (IaC).

2. Architectural Evolution (Minor vs. Major Project)

Layer	Minor Project Baseline	Major Project Enterprise Upgrade
Frontend Console	React + Vite Static UI	React 18 + TypeScript + One-Click Auto-Remediation UI
Backend API	Single-instance FastAPI	Containerized FastAPI Gateway + RBAC + Remediation Engine
Async Processing	Synchronous HTTP execution	Distributed Celery Workers + Redis 7 Message Broker
Database Layer	Local SQLite / Mock State	PostgreSQL 15 + SQLAlchemy Relational ORM
Cloud Remediation	Passive posture reporting	Active Boto3 SOAR Playbooks (S3, EC2 SG, IAM fixes)
Infrastructure	Basic GitHub Pages build	Terraform IaC + Kubernetes (HPA) + Checkov DevSecOps

3. Core Enterprise Modules Implemented

• **Multi-Container Orchestration (Docker & PostgreSQL 15):** Isolates FastAPI, Celery workers, Redis 7, and PostgreSQL 15 on a dedicated bridge network.
• **Asynchronous Worker Queue (Celery & Redis):** Offloads CIS AWS Foundations Benchmark v1.5.0 scans (run_aws_cspm_scan) to background workers.
• **Automated Remediation Engine (Boto3):** Executes programmatic fixes for CIS_AWS_2.1.5 (S3 PublicAccessBlock), CIS_AWS_5.2 (SSH port 22 ingress revocation), and CIS_AWS_1.16 (IAM inline policy detachment).
• **Terraform IaC & Kubernetes HPA:** Provisions AWS VPC networking (10.0.0.0/16) and scales K8s pods from 2 to 6 replicas at 75% CPU threshold.